Linear Algebra and Its Applications

Section 7.2: Quadratic Forms

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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7.2 Quadratic forms

For a symmetric matrix A, the expression $Q(x) = x^T Ax$ is a quadratic form. Diagonal entries are the coefficients of squares. Off-diagonal entries contribute twice: the coefficient of $x_i x_j$ for $i \neq j$ is $2a_{ij}$. Thus put *half* of each mixed-term coefficient in each of the two symmetric positions.

An orthogonal change $x = Py$ gives $Q(x) = y^T(P^T AP)y$. If P consists of orthonormal eigenvectors, then $P^T AP = D$ is diagonal and $Q = \sum_i \lambda_i y_i^2$. All eigenvalues positive means positive definite; all negative means negative definite. Nonnegative eigenvalues with a zero give positive semidefinite but not definite, and the analogous nonpositive case is negative semidefinite. Both signs give an indefinite form. A definite inequality is required only for $x \neq 0$, since every quadratic form has $Q(0) = 0$.

Practice problem 7.2.P1: Recognize positive semidefiniteness spectrally

A symmetric matrix is positive semidefinite exactly when all eigenvalues are nonnegative. In an orthonormal eigenbasis, $Q(x) = \sum_i \lambda_i y_i^2 \geq 0$ if every $\lambda_i \geq 0$. Conversely, if $\lambda_j < 0$, taking $y = e_j$ and $x = Pe_j$ gives $Q(x) = \lambda_j < 0$, contradicting semidefiniteness. This argument also includes zero eigenvalues and the zero matrix.

Exercise 7.2.1: Evaluate a two-variable form

(a) First multiply $Ax = (5x_1 + x_2/3, x_1/3 + x_2)^T$. Dotting with x gives $Q(x) = 5x_1^2 + (2/3)x_1x_2 + x_2^2$.

(b) For $x = (6, 1)^T$, the value is $5(36) + (2/3)6 + 1 = 185$.

(c) For $x = (1, 3)^T$, the value is $5 + (2/3)3 + 9 = 16$.

The two off-diagonal contributions both matter; omitting one changes the answer.

Exercise 7.2.2: Evaluate a three-variable form

(a) Expanding $x^T Ax$ and collecting the symmetric terms gives $Q(x) = 4x_1^2 + 2x_2^2 + x_3^2 + 6x_1x_2 + 2x_2x_3$. There is no x_1x_3 term because $a_{13} = a_{31} = 0$.

(b) At $x = (2, -1, 5)^T$, this is $16 + 2 + 25 - 12 - 10 = 21$.

(c) At $x = (1, 1, 1)^T/\sqrt{3}$, each square and each pair product is $1/3$. Thus $Q(x) = (4+2+1+6+2)/3 = 5$.

Exercise 7.2.3: Recover two symmetric coefficient matrices

The diagonal entries come from the square terms and each mixed coefficient is divided by two. Thus the matrices are

$$\text{(a)} \quad \boxed{\begin{bmatrix} 10 & -3 \\ -3 & -3 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 5 & 3/2 \\ 3/2 & 0 \end{bmatrix}}.$$

For example the first mixed coefficient is $-3 - 3 = -6$, and the second form has no x_2^2 term, so its bottom-right entry is zero.

Exercise 7.2.4: A large mixed coefficient and a pure mixed term

The same rule gives

$$\text{(a)} \quad \boxed{\begin{bmatrix} 20 & 15/2 \\ 15/2 & -10 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}}.$$

The two $1/2$ entries in part (b) sum to the required coefficient 1 of x_1x_2; neither diagonal entry is needed.

Exercise 7.2.5: Three-variable coefficient matrices

Record the three square coefficients on the diagonal, then place half of each of the three possible mixed coefficients in symmetric positions:

$$\text{(a)} \quad \boxed{\begin{bmatrix} 8 & -3 & 2 \\ -3 & 7 & -1 \\ 2 & -1 & -3 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}}.$$

In part (b) all square coefficients vanish, explaining the zero diagonal. Expanding either matrix as x^TAx reproduces every original coefficient.

Exercise 7.2.6: Mixed terms with odd coefficients

Fractions are necessary when a mixed coefficient is odd. The two matrices are

$$\text{(a)} \quad \boxed{\begin{bmatrix} 5 & 5/2 & -3/2 \\ 5/2 & -1 & 0 \\ -3/2 & 0 & 7 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}}.$$

For instance $2(5/2) = 5$ and $2(-3/2) = -3$ give the two mixed coefficients in part (a); the absent x_2x_3 term gives a zero entry.

Exercise 7.2.7: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-4, 6)$ (counted with multiplicity). $\lambda = -4$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_2$. $\lambda = 6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 2)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(-4, 6)$, so

$$Q(x) = \boxed{-4y_1^2 + 6y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.8: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 9 & -4 & 4 \\ -4 & 7 & 0 \\ 4 & 0 & 11 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(3, 9, 15)$ (counted with multiplicity). $\lambda = 3$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_3, \quad x_2 = -2x_3$. $\lambda = 9$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_3}{2}, \quad x_2 = x_3$. $\lambda = 15$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_3, \quad x_2 = -\frac{x_3}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}.$$

Their squared lengths are $(9, 9, 9)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{3}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(3, 9, 15)$, so

$$Q(x) = \boxed{3y_1^2 + 9y_2^2 + 15y_3^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.9: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(2, 7)$ (counted with multiplicity). $\lambda = 2$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 2x_2$. $\lambda = 7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(2, 7)$, so

$$Q(x) = \boxed{2y_1^2 + 7y_2^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.10: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 9 & -4 \\ -4 & 3 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(1, 11)$ (counted with multiplicity). $\lambda = 1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{x_2}{2}$. $\lambda = 11$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(1, 11)$, so

$$Q(x) = \boxed{y_1^2 + 11y_2^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.11: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 2 & 5 \\ 5 & 2 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-3, 7)$ (counted with multiplicity). $\lambda = -3$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_2$. $\lambda = 7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 2)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{2}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(-3, 7)$, so

$$Q(x) = \boxed{-3y_1^2 + 7y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.12: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-6, -1)$ (counted with multiplicity). $\lambda = -6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_2$. $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{x_2}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-6, -1)$, so

$$Q(x) = \boxed{-6y_1^2 - y_2^2}.$$

The form is **negative definite**. Every nonzero y has a nonzero squared coordinate with a negative coefficient, so $Q < 0$. Invertibility of P transfers this to nonzero x.

Exercise 7.2.13: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(0, 10)$ (counted with multiplicity). $\lambda = 0$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_2$. $\lambda = 10$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{3}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}.$$

Their squared lengths are $(10, 10)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{10}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(0, 10)$, so

$$Q(x) = \boxed{10y_2^2}.$$

The form is **positive semidefinite but not positive definite**. All terms are nonnegative, but a nonzero vector along a zero-eigenvalue coordinate gives $Q = 0$, preventing strict positive definiteness.

Exercise 7.2.14: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 8 & 3 \\ 3 & 0 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-1, 9)$ (counted with multiplicity). $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{3}$. $\lambda = 9$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}.$$

Their squared lengths are $(10, 10)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{10}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(-1, 9)$, so

$$Q(x) = \boxed{-y_1^2 + 9y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.15: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} -2 & 2 & 2 & 2 \\ 2 & -6 & 0 & 0 \\ 2 & 0 & -9 & 3 \\ 2 & 0 & 3 & -9 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-12, -8, -6, 0)$ (counted with multiplicity). $\lambda = -12$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = 0, \quad x_3 = -x_4$. $\lambda = -8$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = -6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = -2x_4, \quad x_3 = x_4$. $\lambda = 0$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_4, \quad x_2 = x_4, \quad x_3 = x_4$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & -1 & 2 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 4, 6, 12)$ and their pairwise dot products vanish. Set

$$P = V \operatorname{diag}\left(\frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{6}\right); \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-12, -8, -6, 0)$, so

$$Q(x) = \boxed{-12y_1^2 - 8y_2^2 - 6y_3^2}.$$

The form is **negative semidefinite but not negative definite**. All terms are nonpositive, but a nonzero vector along the zero-eigenvalue coordinate gives $Q = 0$, preventing strict negative definiteness.

Exercise 7.2.16: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 4 & \frac{3}{2} & 0 & -2 \\ \frac{3}{2} & 4 & 2 & 0 \\ 0 & 2 & 4 & \frac{3}{2} \\ -2 & 0 & \frac{3}{2} & 4 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(\frac{3}{2}, \frac{3}{2}, \frac{13}{2}, \frac{13}{2})$ (counted with multiplicity). $\lambda = \frac{3}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} + \frac{5x_4}{4}$, $\quad x_2 = -\frac{5x_3}{4} - \frac{3x_4}{4}$. $\lambda = \frac{13}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} - \frac{5x_4}{4}$, $\quad x_2 = \frac{5x_3}{4} - \frac{3x_4}{4}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 4 & 3 & 4 \\ -5 & 0 & 5 & 0 \\ 4 & -3 & 4 & -3 \\ 0 & 5 & 0 & -5 \end{bmatrix}.$$

Their squared lengths are $(50, 50, 50, 50)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{5\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(\frac{3}{2}, \frac{3}{2}, \frac{13}{2}, \frac{13}{2})$, so

$$Q(x) = \boxed{\frac{3y_1^2}{2} + \frac{3y_2^2}{2} + \frac{13y_3^2}{2} + \frac{13y_4^2}{2}}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.17: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & \frac{9}{2} & 0 & -6 \\ \frac{9}{2} & 1 & 6 & 0 \\ 0 & 6 & 1 & \frac{9}{2} \\ -6 & 0 & \frac{9}{2} & 1 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-\frac{13}{2}, -\frac{13}{2}, \frac{17}{2}, \frac{17}{2})$ (counted with multiplicity). $\lambda = -\frac{13}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} + \frac{5x_4}{4}, \quad x_2 = -\frac{5x_3}{4} - \frac{3x_4}{4}$. $\lambda = \frac{17}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} - \frac{5x_4}{4}, \quad x_2 = \frac{5x_3}{4} - \frac{3x_4}{4}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 4 & 3 & 4 \\ -5 & 0 & 5 & 0 \\ 4 & -3 & 4 & -3 \\ 0 & 5 & 0 & -5 \end{bmatrix}.$$

Their squared lengths are $(50, 50, 50, 50)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{5\sqrt{2}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T A P = \operatorname{diag}(-\frac{13}{2}, -\frac{13}{2}, \frac{17}{2}, \frac{17}{2})$, so

$$Q(x) = \boxed{-\frac{13y_1^2}{2} - \frac{13y_2^2}{2} + \frac{17y_3^2}{2} + \frac{17y_4^2}{2}}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.18: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 11 & -6 & -6 & -6 \\ -6 & -1 & 0 & 0 \\ -6 & 0 & 0 & -1 \\ -6 & 0 & -1 & 0 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-7, -1, 1, 17)$ (counted with multiplicity). $\lambda = -7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = -2x_4, \quad x_3 = x_4$. $\lambda = 1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = 0, \quad x_3 = -x_4$. $\lambda = 17$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -3x_4, \quad x_2 = x_4, \quad x_3 = x_4$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 2 & 0 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 \end{bmatrix}.$$

Their squared lengths are $(4, 6, 2, 12)$ and their pairwise dot products vanish. Set

$$P = V \operatorname{diag}(\frac{1}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{6}); \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-7,-1,1,17)$, so

$$Q(x) = \boxed{-7y_1^2 - y_2^2 + y_3^2 + 17y_4^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.19: A unit-circle maximum

Since $x_1^2 + x_2^2 = 1$, $5x_1^2 + 8x_2^2 = 5 + 3x_2^2 \le 8$. Equality occurs at $x = (0,1)$ or $(0,-1)$. Thus the largest value is $\boxed{8}$. The calculation provides both an upper bound and a point attaining it.

Exercise 7.2.20: A negative coefficient lowers the maximum

Again use the constraint to write $5x_1^2 - 3x_2^2 = 5 - 8x_2^2 \le 5$. Equality occurs at $x = (1,0)$ or $(-1,0)$, so the largest value is $\boxed{5}$. Without the unit constraint, scaling x_1 would make the form arbitrarily large.

Exercise 7.2.21: Six true–false statements about quadratic forms

(a) **True.** The matrix of a form in this section means its canonical symmetric coefficient matrix, with mixed coefficients split equally. A nonsymmetric representing array can be replaced by its symmetric part.

(b) **True.** In a symmetric matrix the mixed coefficient is $2a_{ij}$, so all mixed coefficients vanish exactly when every off-diagonal entry vanishes.

(c) **True.** The principal-axis directions are given by an orthonormal eigenvector basis of the symmetric coefficient matrix.

(d) **False.** The condition is $Q(x) > 0$ for all $x \neq 0$. At $x = 0$ every quadratic form equals zero.

(e) **True.** In eigenvector coordinates the form is a sum of positive coefficients times squares, strictly positive for nonzero inputs.

(f) **True.** This is the stated form of a Cholesky factorization. Its existence is restricted to positive definite symmetric matrices; the statement describes the factorization, not a claim that every symmetric matrix admits one.

Exercise 7.2.22: Five further classification statements

(a) **True.** $\|x\|^2 = x^TIx$ is a quadratic form.

(b) **False.** Orthogonality of P alone is insufficient; its columns must be appropriate eigenvectors. For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $P = I$, the mixed term $2y_1y_2$ remains.

(c) **False.** Degenerate or empty level sets are also possible. For $A = I$, $c = -1$ the set is empty; for $A = \operatorname{diag}(1,0)$, $c = 0$ the level set is the line $x_1 = 0$, not one of the three listed curves.

(d) **False.** An indefinite form takes both positive and negative values. Semidefinite forms have only one weak sign. For example $x_1^2 - x_2^2$ is indefinite and neither kind of semidefinite.

(e) **True.** For each unit eigenvector u_i, $u_i^T A u_i = \lambda_i$. Strict negativity at every nonzero input therefore makes every eigenvalue negative.

Exercise 7.2.23: Compare characteristic-polynomial coefficients

For $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$,

$$\det(A - tI) = (a - t)(d - t) - b^2 = t^2 - (a + d)t + (ad - b^2).$$

On the other hand it is $(t - \lambda_1)(t - \lambda_2) = t^2 - (\lambda_1 + \lambda_2)t + \lambda_1\lambda_2$. Equality of polynomial coefficients gives

$$\boxed{\lambda_1 + \lambda_2 = a + d, \qquad \lambda_1\lambda_2 = \det A}.$$

The factor signs agree because there are two factors; in odd dimensions one must distinguish $\det(A - tI)$ from $\det(tI - A)$.

Exercise 7.2.24: Classify nonsingular two-variable forms without solving for roots

(a) If $\det A = ad - b^2 > 0$ and $a > 0$, then $ad > b^2 \geq 0$, so $d > 0$ and $a + d > 0$. The two real eigenvalues have positive product, so they have the same nonzero sign; their positive sum forces both positive. Thus the form is positive definite.

(b) If $\det A > 0$ and $a < 0$, then $ad > 0$ makes $d < 0$, so the eigenvalues have the same nonzero sign and negative sum. Both are negative, giving a negative definite form.

(c) If $\det A < 0$, the real eigenvalues have negative product, so one is positive and one negative. Their eigenvector directions give opposite values of the form, proving indefiniteness.

The hypothesis $\det A \neq 0$ excludes the semidefinite boundary cases.

Exercise 7.2.25: Gram matrices are positive semidefinite

The matrix $B^T B$ is symmetric. For any x, $x^T B^T B x = (Bx)^T (Bx) = \|Bx\|^2 \geq 0$, proving positive semidefiniteness. If B is square and invertible, $x \neq 0$ implies $Bx \neq 0$, making the last value strictly positive. Thus $B^T B$ is positive definite then. In fact independent columns suffice even for a rectangular B.

Exercise 7.2.26: Construct a positive definite square root

Write $A = PDP^T$ with orthogonal P and all $\lambda_i > 0$ on the diagonal of D. Let $C = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$, so $C^T C = D$, and set $B = PCP^T$. This B is symmetric and, for nonzero x with $y = P^T x \neq 0$, $x^T B x = \sum_i \sqrt{\lambda_i} y_i^2 > 0$, so B is positive definite. Finally

$$B^T B = PC^T P^T PCP^T = PC^T CP^T = PDP^T = A.$$

Both the factorization and the positive definiteness requested of B have now been verified.

Exercise 7.2.27: A sum of positive definite matrices

The sum $A+B$ is symmetric. For $x \neq 0$, $x^T(A+B)x = x^TAx + x^TBx > 0+0 = 0$, since each matrix is positive definite. Thus $A+B$ is positive definite, and its eigenvalues are all positive. The proof does not require A and B to commute or share eigenvectors; adding their eigenvalue lists would not be a valid general argument.

Exercise 7.2.28: The inverse of a positive definite matrix

An orthogonal diagonalization $A = PDP^T$ has positive diagonal entries λ_i. Its inverse is $A^{-1} = PD^{-1}P^T$, with positive eigenvalues $1/\lambda_i$. Hence its quadratic form is positive definite. For a direct check, given $x \neq 0$ let $y = A^{-1}x \neq 0$. Since A is symmetric, $x^TA^{-1}x = (Ay)^Ty = y^TAy > 0$. Thus both the eigenvalue argument and the defining inequality give the claim.